

Artificial Intelligence Concepts (CS2103)

Mid Semester Examination

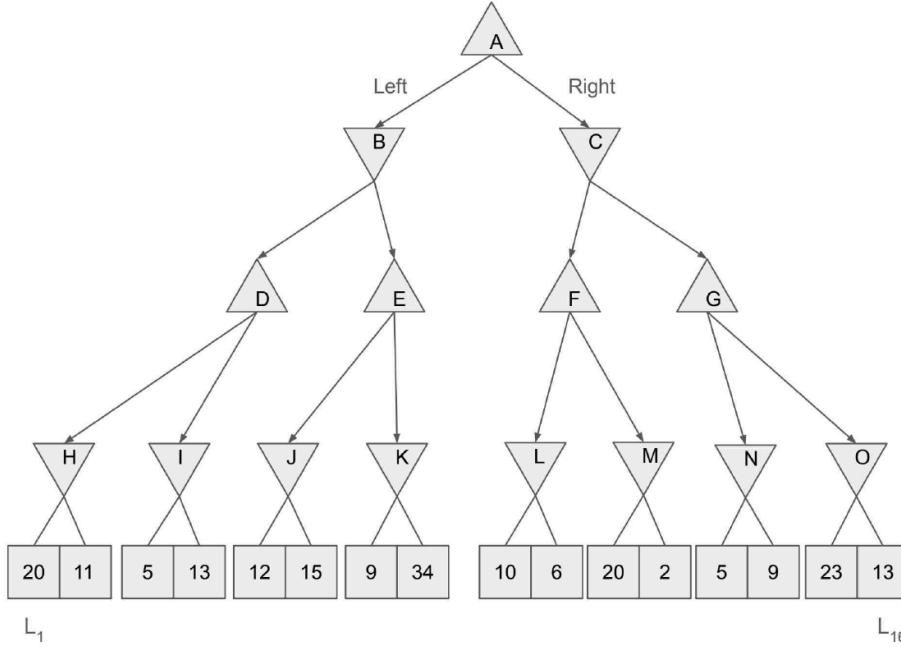
Indian Institute of Technology, Patna

Date: Monday, 23-09-2024

Duration: 2 hrs

Max Marks: 100

[Q1] (12 marks) A two-ply game tree is shown below:



The $\triangle$ nodes are MAX nodes in which it is MAX's turn to move, and the ∇ nodes are MIN nodes. The terminal/leaf nodes are marked L_1 to L_{16} (from left to right) and show the utility values for MAX. The edges represent the actions - LEFT or RIGHT. Assume that the nodes are explored left to right.

Answer the following based on the above stated premise:

[1.1] (3 marks) Find the minimax values for all the non-terminal nodes.

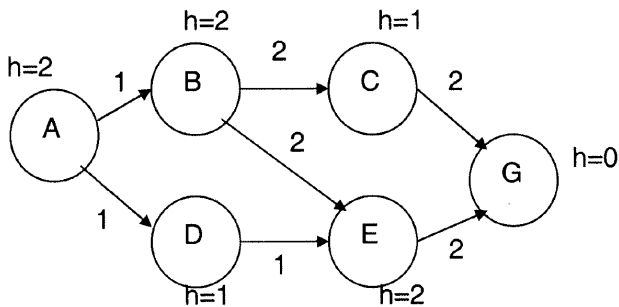
[1.2] (1 mark) What is the optimal action for MAX at the root?

Now, assume that alpha-beta pruning were run on this tree,

[1.3] (4 marks) List the branches that would be cut.

[1.4] (4 marks) List the terminal nodes that won't get explored.

[Q2] (14 marks) The figure shown below shows a problem-space graph, where A is the initial state and G denotes the goal. Edges are labeled with their true cost and the evaluation function is $f(n) = g(n) + h(n)$ where $g(n)$ is the path cost from A to n and $h(n)$ is the heuristics i.e., estimate of best cost to G.



[2.1] (3 marks) Show whether the heuristic is admissible or not.

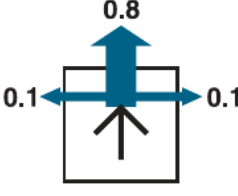
[2.2] (3 marks) Show whether the heuristic is consistent or not.

[2.3] (2 marks) Show whether the evaluation function is monotonic or not.

[2.4] (6 marks) List the nodes visited in order by the A* search. Use alphabetic ordering to break ties.

[Q3] (12 marks) Consider the stochastic (4 x 3) Grid world where the intended outcome occurs with probability 0.8, but with probability 0.2 the agent moves at right angles to the intended direction. A collision with a wall results in no movement. Transitions into the two terminal states (4,3) and (4,2) have reward +1 and -1, respectively, and all other transitions have a reward of $R(s) = -0.04$. Given the current estimate of utilities for states as follows:

3	0.8516	0.9078	0.9578	+1
2	0.8016		0.7003	-1
1	0.7453	0.6953	0.6514	0.4279
	1	2	3	4



(round numerical answers to 6 decimals)

[3.1] (3 marks) Calculate the probability of the agent being in state (3, 1) or (3, 3) after taking the action sequence: Up, Up, Right starting at state (1, 1).

[3.2] (4 marks) Calculate the utilities for states (2, 1) and (3, 2).

Answer the following based on the new utilities calculated in the previous question

[3.3] (2 marks) Find out the optimal actions for states (2, 1) and (4, 1).

[3.4] (3 marks) Suppose that we could change the value of $R(s)$ at this point. What should be the new value of $R(s)$, such that the optimal action for state (4, 1) is Up.

[Q4] (20 marks) Consider a search problem on an infinite 2D-grid of cells where each cell is represented by its integer coordinates (x, y) . Each action has an associated cost of $k > 0$. The agent can take four actions:

1. North: $(x, y) \rightarrow (x, y + 1)$
2. South: $(x, y) \rightarrow (x, y - 1)$
3. East: $(x, y) \rightarrow (x + 1, y)$
4. West: $(x, y) \rightarrow (x - 1, y)$

Assume that the agent starts at cell $(0, 0)$ and the goal state is (p, q) such that $|p| \geq 2$ and $|q| \geq 2$.

[4.1] (2 marks) What is the maximum branch factor (b) and the maximum depth (m) of the search tree?

[4.2] (6 marks) Which of the two search algorithms should be preferred in the given scenario - Breadth First Search (BFS) or Depth First Search (DFS)? Explain why.

[4.3] (4 marks) If $|p| = |q|$ Then what will be the maximum number of nodes expanded by BFS?

[4.4] (3 marks) Assume that we double the cost for actions North and South while keeping the cost for actions East and West the same, will Uniform Cost Search be optimal with this assumption? If so, why?

[4.5] (3 marks) Show that manhattan distance is an admissible heuristics for an A* search algorithm in the given scenario.

[4.6] (2 marks) Show that if manhattan distance is an admissible heuristics then euclidean distance must also be an admissible heuristic.

[Q5] (4 marks) Define the following terms in regards to search strategies:

- [5.1]** Completeness
- [5.2]** Cost Optimality
- [5.3]** Time Complexity
- [5.4]** Space Complexity

[Q6] (14 marks) In the Wumpus World, the environment can be represented using propositional logic. Let's define the following:

- $P(i, j)$: There is a pit in cell (i, j) .
- $W(i, j)$: The Wumpus is in cell (i, j) .
- $B(i, j)$: There is a breeze in the cell (i, j) .
- $S(i, j)$: There is a stench in cell (i, j) .
- $L(i, j)$: Agent is in the cell (i, j) .
- $A(i, j)$: The agent is in a cell adjacent to (i, j) .
- Actions: The agent can move from cell (i, j) to one of its four adjacent cells: $(i+1, j)$, $(i-1, j)$, $(i, j+1)$, $(i, j-1)$.

[6.1] (2 marks) Define a propositional logical rule which says: *If and only if there is a pit in cell (i, j) , then cells adjacent to (i, j) will have a breeze.*

[6.2] (2 marks) Based on the rule defined in the previous question, consider a cell (i, j) where a breeze is perceived by the agent. Write a statement that is equivalent to this perception.

[6.3] (2 marks) Assuming that there is only one cell in the world that contains a stench, define a logical rule which says: *If there is a stench in the cell (i, j) , then the wumpus must be there or nearby.*

[6.4] (2 marks) Assuming that there is only one cell in the world that contains a pit, define a logical rule which says: *The wumpus is never inside the pit or even adjacent to it.*

[6.5] (3 marks) Based on the assumptions and the rules defined till now, given that both breeze and stench are perceived in a cell (i, j) , show if the following statements are entailed or not:

[6.5.1] $P(i+1, j) \vee P(i-1, j) \vee P(i, j+1) \vee P(i, j-1)$

[6.5.2] The wumpus is in one of the adjacent cells.

[6.5.3] The wumpus is not in the cell (i, j) .

[6.6] (3 marks) Assume the following survival criteria for the agent: *The agent cannot survive falling into the pit but it has a survival chance of 25% against the wumpus.* If the agent took a random action in the cell (i, j) where both breeze and stench are perceived, then what is the chance of survival (in %age)?

[Q7] (12 marks) Let Γ and Δ be sets of sentences in Propositional Logic, and let ϕ and ψ be individual sentences in Propositional Logic.

[7.1] (1 marks) Consider the statement: If $\Gamma \models \phi$ and $\Delta \models \phi$, then $\Gamma \cap \Delta \models \phi$
Show that the above statement is false by giving a contradiction.

[7.2] (1 marks) Consider the statement: If $\Gamma \models \psi$, then $\Gamma \models \neg\psi$
Show that the above statement is false by giving a contradiction.

[7.3] (3 marks) The monotonicity theorem says that: If $\Gamma \models \phi$ and $\Gamma \subseteq \Delta$, then $\Delta \models \phi$
Use the monotonicity theorem to show that the following statement is true:
If $\Gamma \models \phi$ and $\Delta \models \phi$, then $\Gamma \cup \Delta \models \phi$

[7.4] (3 marks) Build the truth table for these three sentences:

- (1) $p \rightarrow q \vee r$ (2) $p \rightarrow r$ (3) $q \rightarrow r$

[7.5] (4 marks) Use the truth table built in the previous question to show if the following four logical entailments are correct or not:

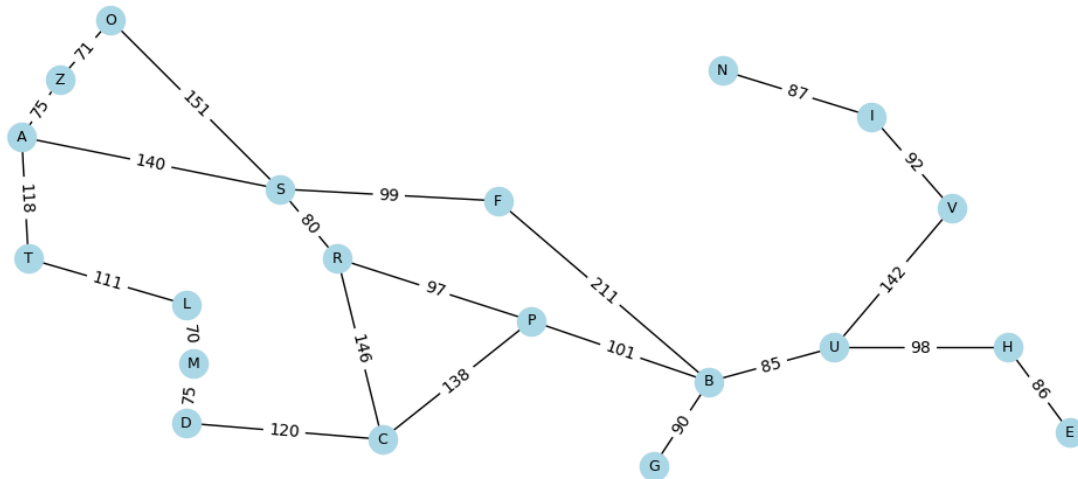
- (1) $\{p \rightarrow q \vee r\} \models (p \rightarrow r)$ (2) $\{p \rightarrow r\} \models (p \rightarrow q \vee r)$
(3) $\{q \rightarrow r\} \models (p \rightarrow q \vee r)$ (4) $\{p \rightarrow q \vee r, q \rightarrow r\} \models (p \rightarrow r)$

[Q8] (12 marks) A search algorithm is defined below in python:

```
def search(graph, start, goal):
    import heapq
    queue = [(0, start, [])] # (cumulative cost, current node, path)
    visited = set()
    while queue:
        .....cost, node, path = heapq.heappop(queue)
        .....if node in visited: continue
        .....path = path + [node]
        .....visited.add(node)
        .....if node == goal: return cost, path
        .....for neighbor, data in graph[node].items():
        .....if neighbor not in visited:
        .....heapq.heappush(queue, (cost + data['weight'], neighbor, path))
    return float("inf"), [] # If no path is found
```

[8.1] (1 mark) Which search method does the given function represent?

[8.2] (8 marks) Given the Romania map (romania_map:networkx.Graph) as shown below:



What will be the output of the following piece of code?

```
cost, path = search(romania_map, "A", "B")
print(cost, path)
```

[8.3] (3 marks) Modify the search algorithm to include heuristics. Assume an arbitrary heuristic function: heuristic(node)->float.